

Optimization of Micro-interconnection Distribution of Gold Stud Bumps for Thermo-sonic Flip Chip Bonding

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Abstract—With the scaling down of broadband RF integration circuits, thermo-sonic flip chip (TSFC) bonding based on Au stud bump has become an important packaging technique for its simpler and more convenient assembly process, compared with solder bump reflow flip chip bonding and anisotropic conductive film flip chip bonding. However, due to the decreasing chip thickness to meet the requirements of IC package and improve the heat conductive properties, the requirements on TSFC bonding parameters are higher, especially for bonding force, to avoid chip fragmentation and reliability problems. In this paper, a numerical model was proposed to avoid stress concentration during TSFC bonding by optimizing the distribution of pads and gold bumps on the chip, to expand the process window and reduce the risk of debris. Finite element software, COMSOL Multiphysics was used to analyze the relationship between surface stress and gold bump spacing. The gold stud bumps used in the simulation are 70 μm in diameter and 50 μm in height. The simulation results indicate that the surface stress distribution is affected by both the number and distribution of the bumps. For the same bonding force, the maximum stress for the chip with one bump is larger than that with two bumps. The stress in the center of two bumps reached the minimum value when the spacing is 260 μm . When the bump spacing is between 260 μm and 300 μm , both the maximum stress and the maximum stress difference on the chip are maintained at lower levels, and the deformation of the chip is relatively small. During the TSFC bonding, the risk of debris can be reduced to a lower level.

Keywords—Gold bump, thermos-sonic, flip chip, stress distribution

I. INTRODUCTION

Gold stud bumps are widely used in wireless interconnection of RF integration circuits for its convenient fabrication process, high bonding strength and low cost [1]. Due to its compatibility of pad materials of GaAs/GaN RF chips, no UBM layers are required [2]. Besides, with no reflow soldering process, solder masks are not required,

leading to more flexible thermal management. The bonding process of gold stud bumps is shown in Fig. 1, which is known as the thermal-sonic flip chip (TSFC) bonding [3]. During the bonding process, the substrate is fixed by vacuum adsorption, while both ultrasonic vibration and pressure are applied to the chip through the bonding tool, and metal bonds are formed by the inter-diffusion of Au atoms. The bonding strength is closely related to the process parameters including ultrasonic power, bonding force, bonding time and temperature, which has been reported in previous researches [4-6].

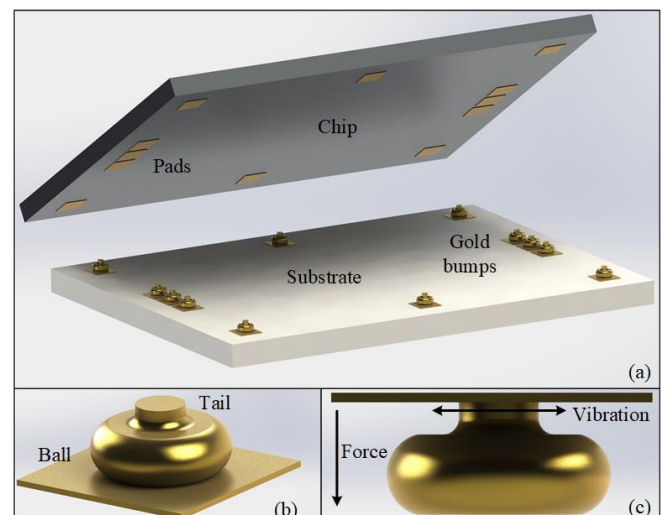


Fig. 1. Diagrams of : (a) TSFC bonding; (b) gold stud bump and (c) physical process of the TSFC bonding.

However, due to the decreasing chip thickness to meet the requirements of packaging and thermal properties, the requirements are higher for the bonding force, to avoid chip fragmentation, false bonding and other reliability problems,

which can be hardly overcome by process parameter optimization.

In this paper, we optimized the micro-interconnection distribution of gold stud bumps to decrease the internal stress, so that the risk of chip fragmentation can be minimized while maintaining high bonding strength.

II. ANALYSIS AND SIMULATION

TSFC bonding is a physical bonding process, in which the role of ultrasonic is to remove the oxides and residues on the bonding surface, and form high-density dislocations on the bonding surface, providing a channel for the subsequent mutual diffusion of Au atoms [7]. Ultrasonic vibration only acts on the surface of the chip pads and will not cause chip deformation and fragmentation without external force. The effect of bonding force is to promote the mutual diffusion of Au atoms at the bonding interface [8]. Overpressure is the main cause of chip fragmentation, while low pressure will lead to the decrease of bonding strength or false bonding [9]. The effects of bonding temperature and bonding time is to promote the diffusion of gold atoms.

Numerical simulations are performed by using the finite element software COMSOL Multiphysics to simulate the influence of bonding force on the chip deformation and fragmentation. The solid mechanics module is used for simulation. Because of its rotational symmetry, the axial sections are selected as the calculation domain, as shown in Fig. 2. The gold stud bumps used in simulation are 70 μ m in diameter and 50 μ m in height.

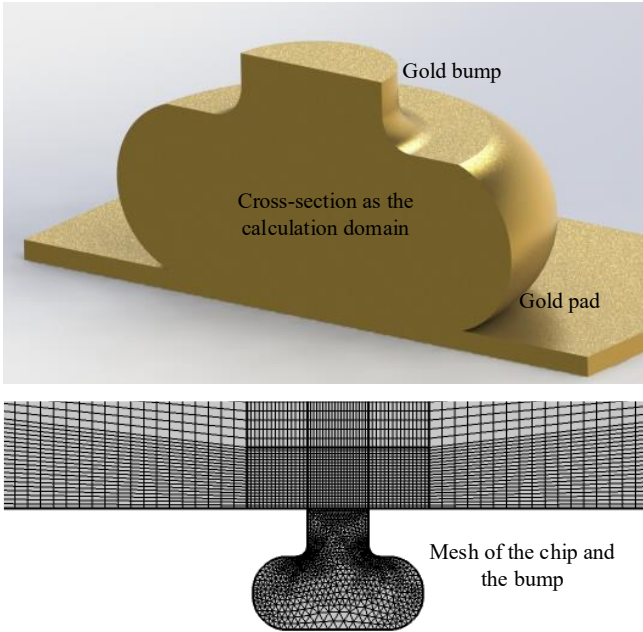


Fig. 2. The geometry of the simulation.

III. RESULTS

A. Simulation of chips deformation

First, numerical calculations were carried out to simulate the deformation of RF chips with different number of bumps and the results are shown in Fig.3. As can be seen in the figure, the deformation differs a lot under the same bonding force. The chip with one bump shows large deformation than that with two bumps, and the three-bump-chip shows the minimum deformation. It follows that more bumps on a chip

under the same bonding force leads to smaller chip deformation. However, the bonding pressure decreases at the same time with the increasing number of bumps, which results in smaller bonding strength.

Therefore, we can come to a conclusion that there is a trade-off between the deformations of the chip and the bonding strength.

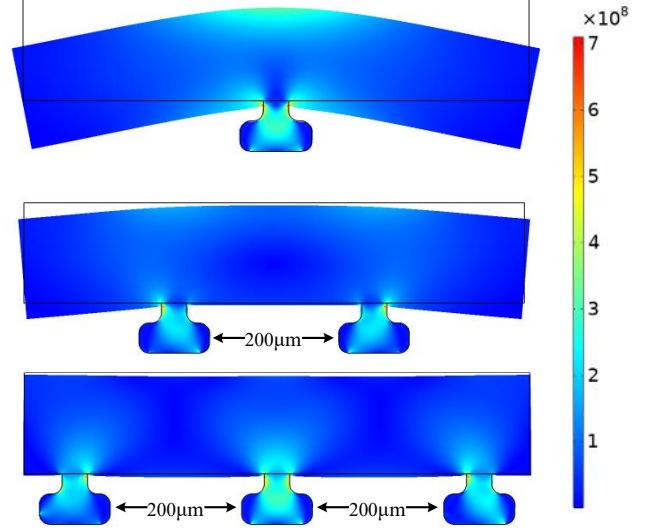


Fig. 3. Deformation of RF chips with different number of bumps.

B. Simulation of bump spacing

Then, the influence of bump spacing on the stress distribution was simulated, and the results are shown in Fig.4, with the 1-bump-chip result as a comparison. It can be seen from the figure that chips with two bumps have at least two stress concentration points and the points occur around the bonding bumps. The difference is that the maximum stress point varies with bump spacing. With the increase of the bump spacing, the maximum stress point gradually changes from the bonding bump to the center of the two bumps.

Therefore, the deformation of the chip resulted from the internal stress also varies with bump spacing.

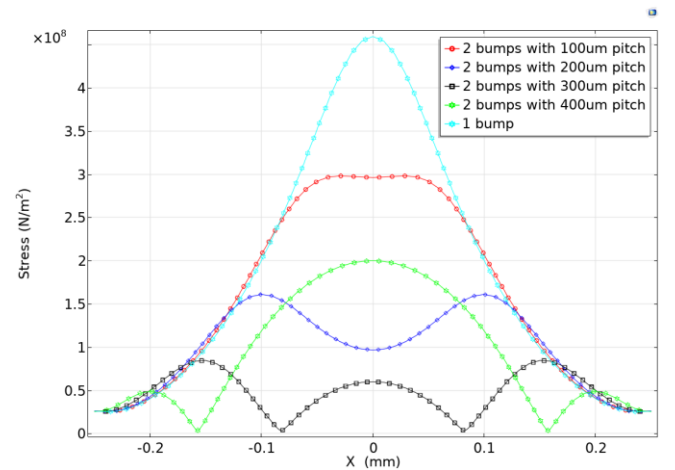


Fig. 4. Simulated stress distribution of chips with different bumps spacing.

C. Numerical analysis of stress concentration

The stress distribution on the chip surface was then simulated with the increase of the bump spacing from 100 μ m

to 400 μm , and the stress at the micro bump and the center of the two bumps were obtained. The results are shown in Fig. 5. As can be seen from the figure, with the increase of bump spacing, the stress at the bump gradually decreases, while the stress at the center of the two bumps, that is, the center of the chip, first decreases and then increases. When the bump spacing reaches 260 μm , the stress at the chip center is almost zero, and there is almost no deformation at the chip center. When the bump spacing is greater than 260 μm , the radial stress component becomes compressive, while it is tensile stress all the time at the bump, thus the deformation of the whole chip will continue to increase.

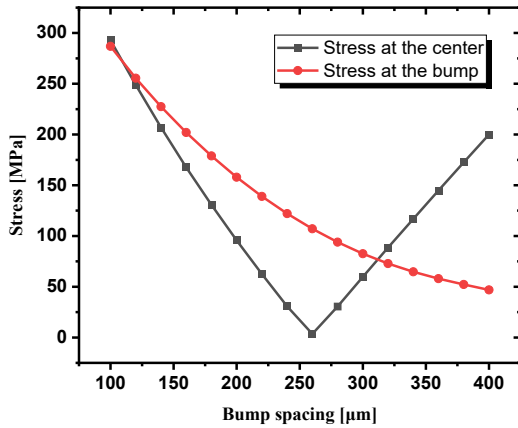


Fig. 5. Simulated stress at the center and the bonding bump with different bump spacing.

The maximum stress, minimum stress and maximum stress difference on the chip surface were further simulated with the increase of the bump spacing, and the results are shown in Figure 6.

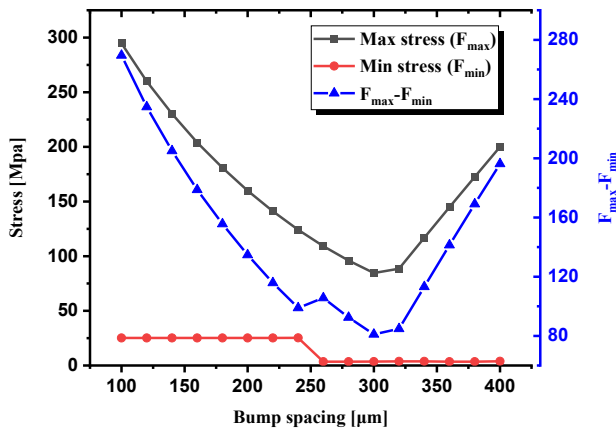


Fig. 6. Variation of the maximum and the minimum stress with the increase of the bump spacing.

It can be seen from the figure that the maximum stress on the chip surface first decreases and then increases with the increase of the bump spacing. When the spacing reaches 300 microns, the maximum stress on the chip surface reaches a minimal value. At this time, the stress at the bump and the stress at the chip center are almost the same, which can be seen from Figure 5. With the increase of the bump spacing, the stress at the chip center exceeds the stress at the bump

location, and becomes the largest stress concentration point. From Fig. 6 we can see that zero stress point appears on the chip surface when the spacing reaches 260 μm . Besides, It can be found that the zero stress point is located in the center of the chip when the spacing is 260 μm , while it moves between the bump and the chip center when the spacing is greater than 260 μm . When the bump spacing is between 260 μm and 300 μm , both the maximum stress and the maximum stress difference on the chip are maintained at lower levels, and the deformation of the chip is relatively small. During the TSFC bonding, the risk of debris can be reduced to a lower level.

IV. CONCLUSION

In this paper, a numerical model was proposed to avoid stress concentration during TSFC bonding by optimizing the distribution of pads and gold bumps on the chip, to expand the process window and reduce the risk of debris.

Firstly, numerical calculations were carried out to simulate the deformation of RF chips with different number of bumps, and we come to a conclusion that there is a trade-off between the deformations of the chip and the bonding strength. More bumps on a chip under the same bonding force leads to smaller chip deformation but weaker bonding strength.

Secondly, the influence of bump spacing on the stress distribution was simulated. We found that with the increase of the bump spacing, the maximum stress point gradually changes from the bonding bump to the center of the two bumps, and the deformation of the chip resulted from the stress also varies with bump spacing.

Thirdly, the stress at the micro bump and the center of the two bumps were simulated with the increase of the bump spacing from 100 to 400 μm . We found that the radial stress component becomes compressive when the bump spacing is greater than 260 μm , while it is tensile stress all the time at the bump. Besides, zero stress point appears on the chip surface when the spacing reaches 260 μm . When the bump spacing is between 260 μm and 300 μm , the deformation of the chip is relatively small, thus the risk of debris can be reduced to a lower level during the TSFC bonding.

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